HARDWARE ENTROPY · PHYSICS OF COMPUTATION

Randomness From Physical Noise

A ring-oscillator true random number generator on the Xilinx Artix-7

Muhammad Saad Bin Waqas *Ghulam Ishaq Khan Institute of Engineering Sciences and Technology*

Nexys A7-100T · XC7A100T-1CSG324C · Vivado 2018.2

> *“The number on the display was not calculated. It was measured.”*

Thermal noise in the silicon, captured through deliberately induced metastability, corrected to remove bias, and confirmed with a real statistical test — using nothing but the FPGA's own logic fabric.

HOW IT WORKS

Every line of code is deterministic — same input, same output, always. To get a value nobody could predict in advance, the design reaches past computation and measures something physical instead: the thermal noise of the chip's own transistors. Five ring oscillators of differing lengths run freely and are XOR-combined; a flip-flop sampling the result on the system clock is deliberately driven into metastability, and which way it resolves is decided by heat, not by logic. No formula is involved at any point in this step.

REMOVING THE BIAS

Physical sources lean slightly one way or another — never a clean 50/50. The Von Neumann corrector reads the stream two bits at a time, keeps only the pairs that disagree, and discards the rest. What survives is unbiased exactly, at a real, quantified cost in speed.

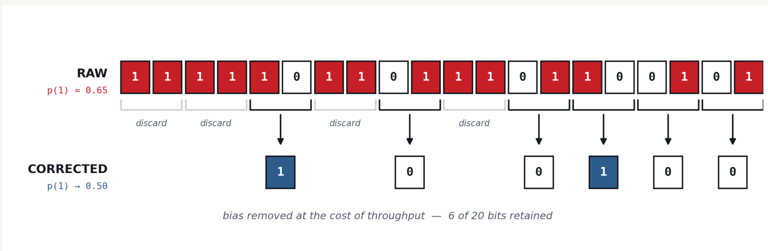

Fig. 1 — Pairwise de-biasing on a deliberately skewed stream.

5	**390.6 k**	**0.500**
RING OSCILLATORS	**RAW SAMPLES / S**	**TARGET OUTPUT BIAS**
3 / 5 / 7 / 9 / 11 stages	100 MHz ÷ 256	guaranteed, not assumed

SCOPE

WHAT THIS SHOWS

- Real physical entropy, captured in FPGA logic alone
- Bias removed exactly, by proof, not by hope
- Output checked with a real statistical test

WHAT THIS ISN'T

- NIST SP 800-90B certified
- Proof the oscillators are independent, not just decorrelated
- A finished, sellable security product

MEASURED RESULTS

full 15-test battery in the accompanying report, Table III

Bits captured	**1,228,280**
Ones / zeros observed	**613,985 / 614,295**
Measured p(1)	**0.4999**
Monobit p-value	**0.7797**
Verdict at α = 0.01	**PASS**
Full NIST SP 800-22 battery	**7 / 15 passed**

TOOLCHAIN Vivado 2018.2 · Verilog-2001 · ~530 lines RTL · Python 3 / pyserial